

Abstract Class

1. Create an abstract class with abstract and non-abstract methods.

Answer :

Animal.java

```
abstract class Animal {  
  
    // Abstract method  
    abstract void sound();  
  
    // Non-abstract method  
    void sleep() {  
        System.out.println("Animal is sleeping.");  
    }  
}
```

2. Create a sub class for an abstract class. Create an object in the child class for the abstract class and access the non-abstract methods

Answer :

Dog.java

```
class Dog extends Animal {  
  
    // Implement abstract method  
    void sound() {  
        System.out.println("Dog barks.");  
    }  
  
    public static void main(String[] args) {  
        // Abstract class reference, Child class object  
        Animal obj = new Dog();  
        // Calling non-abstract method  
        obj.sleep();  
        // Calling abstract method  
        obj.sound();  
  
        // Child class object
```

```

        Dog d = new Dog();

        // Calling abstract method
        d.sound();

        // Calling non-abstract method
        d.sleep();
    }
}

```

Output:

Animal is sleeping.

Dog barks.

Dog barks.

Animal is sleeping.

3. Create an instance for the child class in child class and call abstract methods

Answer:

```

abstract class Animal {

    abstract void sound();
}
class Dog extends Animal {

    void sound() {
        System.out.println("Dog barks.");
    }
    public static void main(String[] args) {

        // Child class object
        Dog d = new Dog();

        // Calling abstract method
        d.sound();
    }
}

```

Output:

Dog barks.

4. Create an instance for the child class in child class and call non-abstract methods

Answer:

```
abstract class Animal {  
  
    void sleep() {  
        System.out.println("Animal is sleeping.");  
    }  
}  
  
class Dog extends Animal {  
  
    public static void main(String[] args) {  
  
        // Child class object  
        Dog d = new Dog();  
  
        // Calling non-abstract method  
        d.sleep();  
    }  
}
```

Output:

Animal is sleeping.